

TensPilot+

An AI-Driven Telemetry and Remote Monitoring Ecosystem for
Electrotherapy



THE UNSEEN GAP IN ELECTROTHERAPY

Traditional TENS therapy suffers from a fundamental flaw in home-care settings: it is an **isolated system**.



Loss of Telemetry

Clinicians cannot verify if or how often therapy is administered.



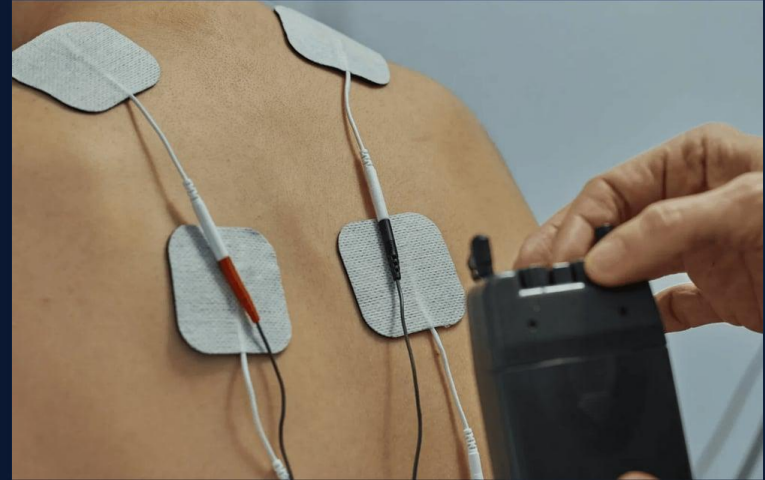
Unmonitored Parameters

No tracking of pulse width, frequency, or intensity.



Qualitative Over Quantitative

Reliance on unreliable paper diaries instead of objective data.



"When telemetry is lost, clinical oversight vanishes."

BRIDGING THE GAP: THE TENSPILOT+ ECOSYSTEM



Patient Node

PWA Companion App built with React & Vite.
Acts as the data-entry and algorithmic logic center.



Clinical Node

Secure Doctor Dashboard for monitoring
aggregated telemetry and compliance curves.



REAL-TIME SYNC



ENGINEERING UNIQUENESS & INNOVATION

🔄 Real-Time Data Propagation

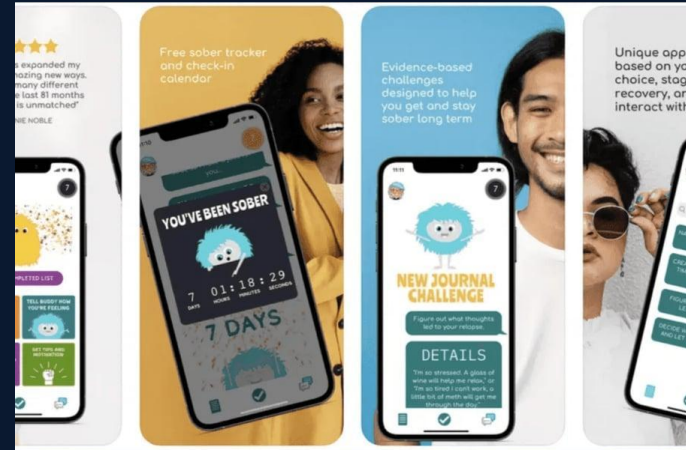
Instant transmission of therapy payloads to Firestore. Doctor dashboards update via reactive snapshot listeners with zero page reloads.

🤖 AI Integration (Gemini API)

A 24/7 virtual physiotherapy assistant providing contextual guidance on pad placement and therapy parameters based on patient queries.

🧠 Algorithmic Optimization

Intelligent decision-support system calculating optimal frequency (Hz) and pulse width (μ s) based on specific pain types (Acute vs. Chronic).



POWERED BY GOOGLE GEMINI & FIREBASE

PRACTICALITY & LOCAL IMPACT



Hardware Agnostic Design

Works with any existing, low-cost TENS machine. No need for expensive proprietary Bluetooth hardware, ensuring maximum accessibility.



Low Bandwidth Optimization

PWA architecture with Service Workers for offline logging. Optimized for low-tier smartphones and intermittent network conditions.



Healthcare System Relief

Reduces physical clinic visits for routine monitoring, allowing clinicians to focus on critical cases in overburdened systems.



Zero Hardware Lock-in

DEMOCRATIZING ACCESS TO TELEMETRY

ADDRESSING SUSTAINABLE DEVELOPMENT GOALS (SDGS)

GOAL 3

Good Health and Well-being

Supports **Universal Health Coverage** by providing remote access to physiotherapy and improving chronic pain management through objective telemetry.

GOAL 5

Gender Equality

Empowers women with **non-pharmacological pain relief** for dysmenorrhea and labor pain, reducing healthcare barriers and improving quality of life.

GOAL 9

Industry, Innovation & Infrastructure

Advances **digital health infrastructure** in developing regions by integrating AI-driven telemetry with existing low-cost hardware.

GOAL 10

Reduced Inequalities

Bridges the gap between urban and rural healthcare access through a **hardware-agnostic, low-bandwidth** design optimized for local constraints.

COMPETITIVE ADVANTAGE: WHY TENSPILOT+ EXCELS

Feature	Traditional TENS	Smart TENS	TensPilot+
Hardware Cost	Low (\$20-\$50)	High (\$150+)	Low (Agnostic)
Clinical Oversight	None (Isolated)	Limited / Local	Full (Real-time)
AI Guidance	None	Basic Presets	24/7 Gemini AI
Connectivity	None	Bluetooth Only	Cloud / Offline



TensPilot+ works with any standard, affordable TENS hardware already in the market.

TAILORED FOR VULNERABLE POPULATIONS



For Nigerians

- **Reduces Travel:** Minimizes expensive trips to urban clinics.
- **Hardware Agnostic:** Works with affordable local TENS units.
- **Low Bandwidth:** Optimized for intermittent connectivity.



For Women

- **Dysmenorrhea:** Discreet, non-drug pain management.
- **Labor Pain:** Adjunct control during childbirth.
- **Post-Op:** Supports recovery after Caesarean sections.

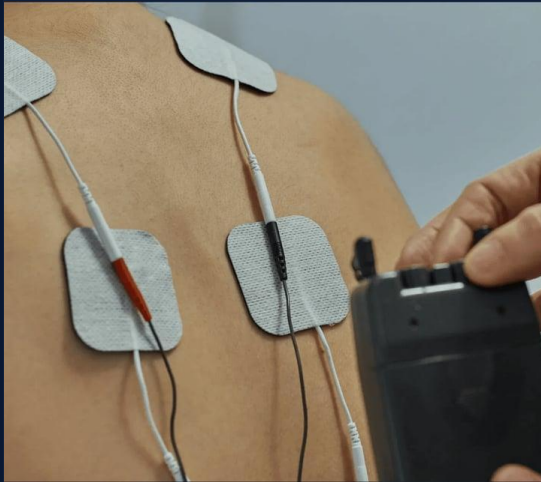


For Children

- **Pediatric Monitoring:** Tracks adherence in musculoskeletal therapy.
- **Objective Data:** Replaces subjective parent reporting.
- **Home-Based:** Less disruptive to school and family life.

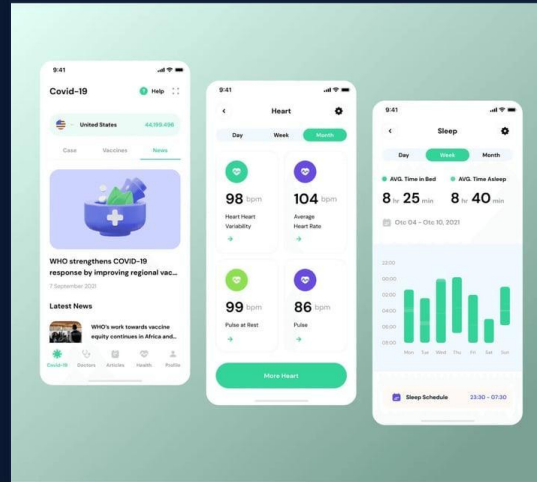


HARDWARE & SOFTWARE IN ACTION



THE HARDWARE

Compatible with standard, low-cost TENS units already available in the local market.



PATIENT PWA

Intuitive mobile interface for session logging, AI guidance, and parameter optimization.



CLINICAL PORTAL

Real-time monitoring dashboard for clinicians to track compliance and patient progress.

Revolutionizing Electrotherapy for a Healthier Future



Improved Outcomes

Enhanced patient adherence through intelligent guidance and real-time feedback.



Clinical Efficiency

Optimized oversight with objective data, reducing the burden on healthcare systems.



Democratized Access

Hardware-agnostic design bringing advanced telemetry to underserved communities.

Join us in transforming pain management.